Paper 72 — SASSHA

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Sharpness-aware adaptive second-order optimization with stable Hessian approximation

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The paper in one minute

Question: Can approximate second-order optimizers retain fast, curvature-aware steps without repeatedly settling in sharp solutions that generalize poorly?
Model/Method: Combine a SAM-like worst-neighborhood gradient with an estimated diagonal Hessian; stabilize the denominator using square roots and reuse curvature estimates.
Finding: Across tested workloads SASSHA finds flatter solutions with competitive or improved validation performance. Its convergence theorem is conditional on *convexity*, not a global theory for neural networks.
Why we care: It distinguishes improving an optimizer's practical generalization from proving saddle escape or global non-convex convergence.

1. First principles: two objectives conflict

A second-order-style method rescales a gradient by curvature, roughly $x^+ = x - \eta H^{-1} g$. This can accelerate poorly conditioned directions, but a small estimated Hessian entry can yield a very large step. Meanwhile, SAM changes which gradients are followed by first looking for the highest local loss inside a ball:

$$\min_x \max_{\|\epsilon\| \le \rho} f(x + \epsilon).$$

A first-order approximation chooses $\epsilon^* = \rho g / \|g\|$. SASSHA combines these ideas: use the perturbed-point gradient for sharpness awareness, but precondition it with inexpensive curvature information.

2. Worked example: why naive inverse curvature can explode

Imagine a two-dimensional parameter vector with gradient $g = (1, 1)^\top$ and diagonal curvature estimate

$$\widehat{H} = \begin{bmatrix} 0.01 & 0 \\ 0 & 4 \end{bmatrix}.$$

An unregularized inverse-Hessian direction is

$$\widehat{H}^{-1} g = (100, 0.25)^\top.$$

The first coordinate is enormous solely because its estimated curvature is tiny. A square-rooted absolute diagonal gives instead

$$|\widehat{H}|^{-1/2} g = (10, 0.5)^\top.$$

The problematic amplification drops from 100 to 10 while retaining the fact that the first direction is flatter. This is an *illustrative scaling calculation*, not the exact SASSHA full update with moving averages, bias correction, and weight decay.

Why absolute values and square roots?
Diagonal curvature estimates can be noisy or negative. Taking their absolute values enables a nonnegative scale; taking a square root softens division by nearly zero estimates. It is not the same as inverting the entire true Hessian.

3. How the algorithm is assembled

At step t, SASSHA uses a minibatch gradient $g_t = \nabla f_{\mathcal{B}}(x_t)$, computes the SAM-style perturbation, and evaluates a second gradient:

$$\epsilon_t^* = \rho \frac{g_t}{\|g_t\|}, \qquad \widetilde{g}_t = \nabla f_{\mathcal{B}}(x_t + \epsilon_t^*).$$

It forms an exponential moving average m_t of $\widetilde{g}_t$, and every k steps estimates the perturbed-point Hessian diagonal. A running curvature statistic D_t tracks *absolute* Hessian entries and the denominator is proportional to $\sqrt{D_t}$, rather than D_t. Between curvature-refresh steps, the old denominator is reused. Algorithm 1 gives the full form, including bias correction and weight decay.

Why would lazy Hessian updates help?

Computing even a diagonal Hessian approximation repeatedly adds cost. The authors hypothesize that sharpness-aware training visits regions where curvature changes more gradually, so a recently computed estimate remains useful longer. This is a practical hypothesis supported by ablations, not a universal identity about curvature evolution.

4. What the mathematical results really guarantee

Theorem 4.4 is important to cite accurately. It assumes a lower-bounded, twice-differentiable *convex*, smooth objective, along with a nonvanishing-gradient technical condition over the specified finite steps and step/radius sequences satisfying

$$\sum_t \eta_t = \infty, \qquad \sum_t \eta_t^2 < \infty, \qquad \sum_t \rho_t^2 \eta_t < \infty.$$

Its conclusion is

$$\boxed{\liminf_{t\to\infty} \|\nabla f(x_t)\| = 0.}$$

This is weaker than asserting convergence of the whole sequence to a unique solution, and it is *not* a general non-convex strict-saddle-escape result. Corollary 4.6 separately studies necessary linear-stability conditions for minima under an idealized stochastic-Hessian model.

Don't overstate the theorem
Good experimental deep-learning performance and a convex convergence theorem are different pieces of evidence. Neither by itself establishes global optimality for arbitrary ReLU networks.

5. Experimental results and contributions

The authors compare SGD, AdaHessian, Sophia-H, Shampoo, and SASSHA using multiple sharpness metrics, loss-surface projections, validation accuracy, stability tests, and wall-clock efficiency. On ResNet-32 / CIFAR-100, their Table 1 reports SASSHA validation accuracy 72.14% versus 69.32% for SGD, with smaller measured maximum Hessian eigenvalue (107 versus 265). These particular numbers are *experimental*, not a universal accuracy improvement. They also study vision and language workloads. **Contributions.** (i) Put SAM-style neighborhood gradients inside an approximate second-order optimizer; (ii) stabilize the curvature denominator via the square root of absolute diagonal estimates; (iii) reuse Hessian approximations at reduced cost; (iv) examine flatness and stability of selected solutions.

6. Assumptions and limitations

- The formal convergence statement uses convexity and carefully decaying step/radius sequences; deep-network training is generally non-convex.
- Diagonal Hessian approximations omit off-diagonal curvature and may be biased/noisy; curvature-refresh intervals introduce a practical tuning trade-off.
- Flatness and validation performance are empirically associated in these experiments, not proved equivalent for all parameterizations.
- There is no general second-order stationary-point or saddle-escape complexity guarantee in this paper.

7. Quick reference

Sharpness objective	$\min_x \max_{\|\epsilon\|\le\rho} f(x+\epsilon)$.
Perturbation	$\epsilon^* = \rho g/\|g\|$.
Gradient used	$\widetilde{g} = \nabla f(x+\epsilon^*)$.
Stable preconditioner	Estimate absolute diagonal Hessian, then use a square-root denominator.
Lazy update	Recompute curvature every k steps; reuse otherwise.
Guarantee	Theorem 4.4: $\liminf \|\nabla f\| = 0$ under convex assumptions.

Survey bridge / Papers Master note

SASSHA combines SAM-style sharpness control and approximate second-order preconditioning, using square-rooted absolute diagonal Hessians and lazy updates for stability and speed. Reports flatter, better-generalizing solutions in studied tasks, but its convergence theory assumes convexity; cite as a practical intervention, not a strict-saddle escape theorem.